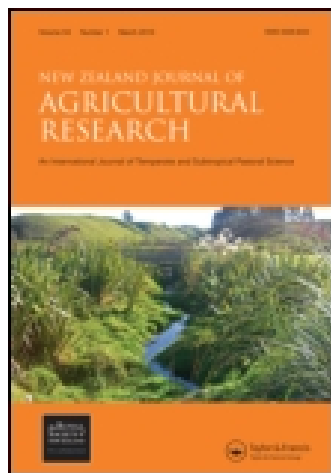


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Uptake of potassium and nitrogen by pasture from urine-affected soil

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Abstract A field experiment was carried out to determine the depth of soil from which pasture plants growing in urine-affected soil recover urine potassium (K) and nitrogen (N). A comparison of the amounts of K and N taken up by the herbage with the decline in soil K and N levels showed that K and N were absorbed mainly from the 0–12 cm depth of soil. This coincides with the distribution of plant roots within the soil profile, as 86% of the roots in the 0–30 cm depth of soil were found in the top 15 cm. The experiment also showed that immediately after a dairy cow urination event in mid winter (July), up to 59% of the urine K moved preferentially through a network of soil macropores to below a soil depth of 14.5 cm. A simulation of the field trial in the glasshouse confirmed that preferential flow of the urine could result in large losses of urine K. Forty-eight percent of the urine-N was also unaccounted for by soil analysis immediately after the urination event, and is considered to have been lost preferentially with the K. The urine K and N that remained in the 0–14.5 cm soil profile as exchangeable K and potassium chloride-extractable N were absorbed by the pasture plants over the following spring and summer.

Keywords potassium; nitrogen; potassium leaching; nitrogen leaching; herbage potassium; herbage nitrogen; plant uptake; urine; K cycle; N cycle

INTRODUCTION

The return of potassium (K) in urine is a major pathway in the K cycle of a grazed dairy pasture. The amount of K deposited in one urine spot is equivalent to an application of c. 1000 kg K/ha (Saunders 1984) which is excessive compared with the 60–70 kg K/ha that is usually applied in fertiliser (During 1984). Significant quantities of nitrogen (N) are also recycled annually in the form of urine and large amounts of N are found in urine spots (c. 500 kg N/ha; Saunders 1984). If these high amounts of K and N are not recovered by the plants they may be subject to loss by leaching. Therefore, the ability of pasture plants to re-utilise the K and N from urine-affected soil will have a major influence on the efficiency of the K and N cycles in the grazed dairy pasture.

A characteristic of most pastoral systems is that the majority of the plant roots are found in the surface layers of the soil profile (Caradus & Evans 1977). The amount of K taken up by the roots of pasture species is related to the proportion of the root mass that is present at any one depth (Ozanne et al. 1965) and it has been shown in field experiments that the majority of the K taken up by pasture plants such as ryegrass and lucerne is derived from the surface layers of the soil profile (Ozanne et al. 1965; Peterson & Smith 1973). In New Zealand, measurements of the decline in soil exchangeable K over time have been used to suggest that pasture plants are capable of recovering K from depths down to 61 cm (Weeda 1978; During & Campkin 1980). However, no measurements have been made on the depths of soil from which pasture plants recover K and N following a urination event. This paper reports on an experiment designed to determine the amounts of K and N that are taken up from various depths of urine-affected soil.

MATERIALS AND METHODS

Trial establishment

The trial site was on established white clover/ryegrass pasture on the Massey University No. 4 dairy farm. The soil was a Tokomaru silt loam (Typic Fragiaqualf) and some of its properties are given in Table 1.

In June 1987, an area (5×3 m) was fenced-off from stock. The pasture was mown to a height of 2 cm above ground-level and 21 cylinders (15 cm in diam. and depth) were hammered into the soil to isolate each soil "core". A fortnight later (8 July 1987) the herbage on each core was trimmed to 2 cm and 200 ml of dairy cow urine was applied to each core to simulate a urination event of 11 litres/m² (Hogg 1981). The urine had been previously collected from the Massey University No. 3 dairy farm during a spring milking, frozen immediately, and stored until required. The concentrations of K and N in the urine were 6.6 and 5.5 mg/ml, respectively. The soil moisture content of the soil when the urine was applied was considered to be equivalent to "field capacity".

Sampling and analysis of herbage and soil

At intervals of c. 5 weeks, the herbage on all the cores was harvested, dried at 65°C for 24 h, weighed, and ground finely. The K concentration of the herbage was determined after digestion of 0.1 g of ground plant material with 4 ml of nitric acid at 150°C for 4 h. The excess liquid was evaporated at 200°C and the residue dissolved in 5 ml of 2M hydrochloric acid. The K concentration in this solution was determined by atomic emission in the presence of 0.2% caesium chloride. The herbage was also

digested with kjeldahl mixture (0.1 g with 5 ml of kjeldahl mixture) for 3 h at 350°C, then the NH_4^+ -N concentration was measured by autoanalyser (Technicon 1973).

After the herbage had been harvested, three cores were retrieved and removed to the laboratory. The soil cores were dried in the oven at 65°C for 1–3 days depending on their moisture content, after which time it was possible to remove the cylinder from around the soil core. The cores were then sliced into three depths (0–2.5, 2.5–12, and 12–14.5 cm) which were left to dry at room temperature. After sieving (<2 mm; any root material that did not pass through the sieve was finely ground and added to the soil), 4 g of soil were shaken twice with 20 ml of 1M ammonium acetate for 30 min to extract exchangeable K (Thomas 1982). The two filtered extracts were combined together and the K concentration measured as before. A further 2 g of soil were shaken with 20 ml of 2M potassium chloride (KCl) for 2 h to extract NH_4^+ -N and NO_3^- -N (Adamsen et al. 1985). The NH_4^+ -N and NO_3^- -N concentrations in the centrifuged, filtered extracts were measured by autoanalyser (Technicon 1976; Downes 1978).

The exchangeable K and KCl-extractable N contents of the soil before the urine application were determined from a sample of soil collected from around the outside of the soil cores. Measurements of soil K and N immediately after the urine application were carried out on three soil cores which were retrieved 3 h after the urine was applied.

Measurement of the distribution of pasture root mass with depth

The distribution of pasture root mass with soil depth on the Tokomaru silt loam was measured adjacent to the field trial area. At 10 sites selected at random throughout the paddock a pit was dug in the soil to a depth of 30 cm. Soil samples of a known volume (491 cm³) were collected at 5 cm depths down to 30 cm. The samples were washed with water to separate the roots from the soil, then the roots were dried at 65°C for 24 h and weighed.

Glasshouse experiment

Intact soil cores of the same dimensions and soil water content as used in the field experiment were collected from the pasture trial area. Each core was placed on top of a plastic container so that leachates could be collected, and all the cores were kept in a glasshouse for the duration of the experiment. Two hundred ml of dairy cow urine was applied to each core. The leachate was collected, weighed, and the

Table 1 Background information on the Tokomaru silt loam.

NZ soil group	Yellow-grey earth
Soil taxonomic classification	Typic Fragiaqualf
pH ^a	6.6
Cation exchange capacity ^b (me/100g)	22
Organic carbon ^c (%)	5.2
Dominant clay minerals ^d	Layered chlorites Vermiculite

^a In water (1:2.5 ratio). ^b Hesse (1971). ^c Tabatabai & Bremner (1970). ^d Pollok (1975).

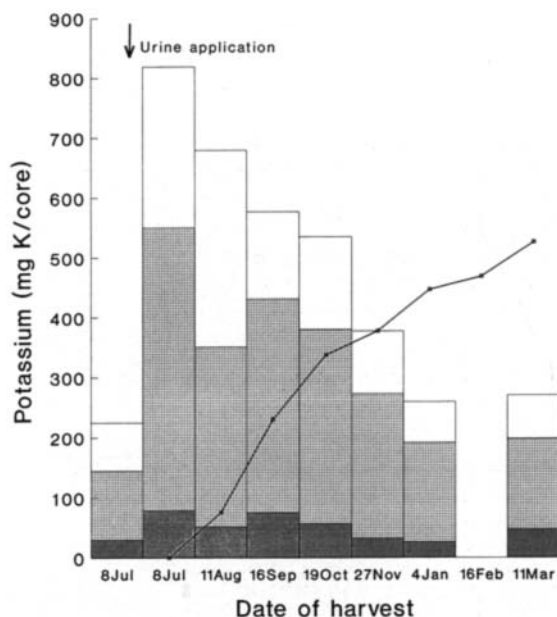


Fig. 1 Soil exchangeable K in the 0–2.5 cm (□), 2.5–12 cm (▨), and 12–14.5 cm (■) depth of soil and cumulative herbage K (—●—) following application of 1320 mg K/core in the form of urine. Values are means of three replicates (CV = 5–20%).

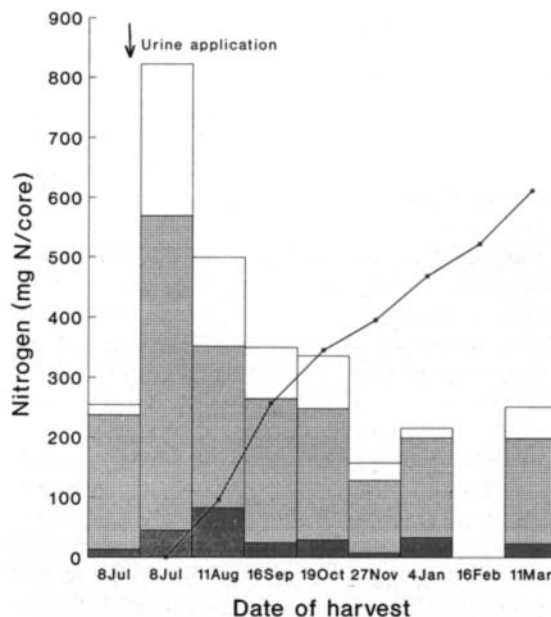


Fig. 2 Sum of KCl-extractable soil $\text{NH}_4^+\text{-N}$ and $\text{NO}_3^-\text{-N}$ in the 0–2.5 cm (□), 2.5–12 cm (▨), and 12–14.5 cm (■) depth of soil and cumulative herbage N (—●—) following application of 1100 mg N/core in the form of urine. Values are means of three replicates (CV = 5–20%).

K concentration measured by atomic emission in the presence of 0.2% caesium chloride. The soil cores were watered daily to “core capacity” by applying deionised water from a washbottle. The daily watering events did not initiate leaching. After 1 month the heavy rain storm events which fell on the field trial in October 1987 were simulated to determine if there were significant quantities of K leached during this period. The simulated rain (between 5 and 23 mm per day for 6 days) was applied in events of 5 mm as 90 ml of deionised water from a washbottle. The weight of the leachate from each 5 mm rain event was determined and the K concentration measured as previously described.

RESULTS AND DISCUSSION

Retention in the soil of urine potassium and nitrogen

The application of urine to the soil increased the amount of exchangeable K in the soil immediately, particularly in the 0–2.5 and 2.5–12 cm depths (Fig. 1). The increase in exchangeable K accounted for

only 41% of the K applied in urine, suggesting that a major loss of K occurred during the urine application.

The data on soil N showed a similar pattern to exchangeable K. There was an immediate increase in KCl-extractable N (mainly $\text{NH}_4^+\text{-N}$ with <6% as $\text{NO}_3^-\text{-N}$) after application of the urine, and the increase was greatest in the 0–2.5 and 2.5–12 cm depths (Fig. 2). The increase in KCl-extractable N accounted for 52% of the urine N applied. Although it was possible that some of the urine N may have been present in the soil as urea at the time of sampling, it was expected that by the time the soil had been dried, sieved, and analysed (c. 7 days) the urea would have been converted to $\text{NH}_4^+\text{-N}$ or $\text{NO}_3^-\text{-N}$. It is considered that the majority of the N not accounted for probably moved through the soil profile during preferential flow of the urine. Experiments with undisturbed soil cores have shown that during preferential flow of urine similar proportions of K and N are leached (Williams 1988).

The results from the glasshouse experiment showed that immediately after the urine was applied to the soil cores, a large amount of K (equivalent to

38–55% of that applied) was collected in the leachate. The volume of urine applied (200 ml) represented 0.15 pore volumes, which would have been insufficient for the urine/soil water dispersion front to reach the bottom of the core. The concentration of K in the leachate (2400–3500 µg/ml) more closely resembled the K concentration of the cow urine (6600 µg/ml) than that usually found in soil solution (3–30 µg/ml), suggesting that the K in the leachate was derived from the preferential movement of urine through the soil core via a network of macropores in the soil. A simulation of the application of urine to the soil cores using methylene blue dye (0.1% solution) confirmed that liquids could move through the soil core via macropores created by plant roots, earthworms, and soil fracture planes. Preferential movement of solutes through intact soil cores of the Tokomaru silt loam has been measured previously by Kanchanasut et al. (1978). The dye study also indicated that the urine moved through the macropores of the soil rather than between the outside of the soil core and the cylinder. The occurrence of edge flow can significantly increase the amount of applied solutes that are collected in the leachate (Cameron et al. 1989).

Losses of K as a result of preferential flow of urine through the soil have not been measured before. The data from the glasshouse study showed that the proportion of applied K that was lost from the soil/plant system via preferential flow was far higher than that reported as being lost via leaching by other authors (Saunders & Metson 1959; Hogg 1968, 1981). Therefore, there is a need to study the preferential flow of urine as an important loss mechanism for nutrients on dairy farms.

Uptake of potassium and nitrogen by plants

The total amount of K that was taken up by the plants in the field experiment was 525 mg/core (Fig. 1). This amount of K is similar to both the increase in the amount of soil exchangeable K brought about by the urine application (580 mg K/core; Fig. 1) and the difference between the total amount of soil exchangeable K at the beginning (after the urine application) and end of the experiment (550 mg K/core; Fig. 1). It is possible that some of the urine K may have been retained by the soil clay minerals in a non-exchangeable form that is not extracted by ammonium acetate. However, other experiments investigating recovery of urine K from this soil type have suggested that retention in this form is small (Williams 1988). Results from the glasshouse study showed that only a small amount of K (27.0 mg K/

core, SEM = 4.3) was leached from the simulated rain events, suggesting that leaching losses of urine K were small. Therefore, the amount of K taken up by the herbage during the field experiment was equivalent to the amount of K retained in the soil as exchangeable K after the urine event.

The amount of N taken up by the herbage (610 mg N/core; Fig. 2) during the 8-month field experiment was comparable with the amount of urine N retained in the 0–15 cm depth of soil after the urine event (570 mg N/core; Fig. 2). Although some of the N taken up by the plants may have been derived from sources of N other than the urine (e.g., mineralisation of soil organic N and fixation by clover plants), it appears that the amount of N taken up by the herbage was equivalent to the amount of N retained in the soil as KCl-extractable N after the urine event.

The amounts of K and N taken up by the herbage in this experiment were equivalent to 40% of the K and 55% of the N applied in the urine. These apparent recoveries of K and N agree with the ranges reported by Saunders & Metson (1959), During & McNaught (1961), Ball & Keeney (1981), Hogg (1981), Carran et al. (1982), and Ledgard & Saunders (1982).

Depth of uptake

Examination of the data in Fig. 1 and 2 in more detail shows that the added urine K and N increased the amounts of exchangeable K and KCl-extractable N in mainly the 0–2.5 and 2.5–12 cm soil depths. Throughout the trial period the quantity of soil K and N in the two top soil depths decreased by 200 and 320 mg K/core, and 200 and 350 mg N/core, respectively. The corresponding decreases in the 12–14.5 cm depth were only 30 mg K/core and 20 mg N/core. The pattern of these decreases indicates that most of the plant K and N was derived from the upper depths of soil with very little uptake of K from below 12 cm. This coincides with the distribution of

Table 2 Distribution of pasture root mass in the Tokomaru silt loam.

Depth of soil (cm)	kg DM/ha	Root weight SEM	%
0–5	4340	331	53
5–10	1850	155	23
10–15	820	86	10
15–20	460	107	6
20–25	410	63	5
25–30	240	40	3
			100%

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plant roots within the soil profile. The majority of the weight of plant roots in the top 15 cm of the soil profile was found in the 0–5 (62%) and 5–10 cm depths (26%; Table 2). Such a result confirms the general finding that plants recover the majority of their nutrient requirement from the surface layers of the soil profile (e.g., Gillingham et al. 1980; Horne 1985).

Another factor which may have resulted in lower uptake from depths greater than 12 cm in this experiment was that this study measured plant uptake from urine-affected soil. Although K has little influence on root growth, added N can increase root density (Barber 1984). Therefore, localised increases in root density as a result of urine N may have enabled the plants to extract their K and N requirements from the 0–12 cm depth without the need for deeper extraction.

The moisture content of the soil can affect the depth of soil from which nutrients can be absorbed by plant roots. The roots of perennial ryegrass swards tend to absorb nutrients from soil depths below 18 cm during periods when the top 5 cm of soil dries out (Garwood & Williams 1967; Newbould et al. 1971). During these periods the roots in the top soil can become inactive (Garwood & Williams 1967). In the field experiment reported in this paper, the surface layers of soil dried out only in January when climatic conditions were very dry (i.e., only 5.2 mm of rain was recorded over 34 days). Because of the dry conditions, herbage growth over this period was very small and this, combined with a low concentration of K and N in the herbage (1.6% K in January compared with 3.8% in August, and 3.6% N in January compared with 4.9% in August) resulted in uptake that was only 5% of the total K and 9% of the total N recovered over the whole trial period. Such a result confirms that although plants may

absorb water from deeper down the soil profile for survival during a dry period, the amount of K and N absorbed from urine-affected soil will be small compared with the total amount recovered during the year.

The decrease in the amounts of soil exchangeable K at each soil depth during the field experiment has been expressed per mg of root weight in each soil depth (Table 3). The amount of K taken up by the plants from each soil depth was considered to be the decrease in the amount of soil exchangeable K during the experimental period. The root weights at each depth (0–2.5, 2.5–12, and 12–14.5 cm) were interpolated graphically using the data in Table 2. The ratios of mg K taken up by plants : mg of root weight were similar for each soil depth (Table 3), which emphasises the known dependence of plant K uptake on the amount of roots present (Barber 1984). Of the total weight of roots in the soil profile only a small proportion was found below a depth of 15 cm (Table 2). Plant-available soil K moves by diffusion over distances of 0.1–15 mm (Barber 1984). Below the 15 cm depth of soil (where root density is low), the amount of K that is likely to diffuse to plant roots must be small compared with that in the upper depths of the soil (where root density is high). Again it is concluded that the amount of K that is taken up from soil depths below 15 cm must be only a small proportion of the total plant K uptake.

Extrapolation of the results of this experiment to other geographical areas is limited to situations where plant roots are concentrated in the topsoil (i.e., >50% of plant roots in the 0–5 cm depth) and the rainfall distribution ensures that the soil surface dries out for only short periods of time. Such areas are likely to be the western areas of New Zealand (e.g., Taranaki, Waikato, and Manawatu) which have summer rainfall.

CONCLUSIONS

The amounts of K and N recovered by the herbage over the 8-month trial period were equivalent to the amounts of exchangeable K and KCl-extractable N retained by the soil after the urination event. Thus it appears that the pasture plants were capable of recovering the urine K and N that was retained by the 0–14.5 cm depth of soil following a urination event in mid winter (July).

The majority of the N and K taken up by the plant roots growing in urine-affected soil was derived from the 0–12 cm depth in this experiment. The decrease in the amount of exchangeable K and KCl-

Table 3 Ratio of the amount of K absorbed by plant roots (as measured by the change in soil exchangeable K during the experiment) to the weight of roots at each soil depth.

Depth of soil (cm)		Change in soil K (mg K/core)	Weight of roots (mg/core)	Ratio
0–2.5		200	4710	0.042
	SEM	12	359	0.007
2.5–12		320	6820	0.047
	SEM	29	571	0.010
12–14.5		30	868	0.035
	SEM	6	91	0.01

extractable N at each soil depth throughout the experiment correlated well with the root density at each soil depth. Although some K and N may be taken up from further down the soil profile, particularly during dry periods, the amounts are likely to be small compared with total recovery for the year.

The amounts recovered by plant uptake were equivalent to only 40 and 55% of the K and N applied in the urine. For a typical New Zealand dairy farm with a stocking rate of 3 cows/ha there are c. 230 kg K/ha and 180 kg N/ha recycled annually to the pasture in the form of urine. If pasture plants are only capable of recovering 40–55% of these amounts of K and N, then there are significant quantities of urine K and N in the soil which are at risk of being lost from the grazed pasture. The extent of the soil loss has a direct effect on fertiliser requirements for these nutrients, particularly K. A large proportion of the urine K and N unaccounted for may have been lost during the urine application. Immediately following a urination event in winter onto ryegrass/white clover pasture, 41% of the K and 52% of the N applied in urine were retained in the 0–14.5 cm depth of the Tokomaru silt loam as exchangeable K and KCl-extractable N, respectively. A simulation of the field experiment in the glasshouse indicated that the remainder of the urine K was lost by immediate preferential flow of urine to lower soil depths. A similar fate is thought to have occurred with the urine N that was unaccounted for by soil analysis. This mechanism represents a major potential loss of K and N from the grazed dairy pasture.

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